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"Dr. AI Will See You Now": How Do ChatGPT-4 Treatment Recommendations Align With Orthopaedic Clinical Practice Guidelines?

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CLINICAL & AGENTIC AI TAKEAWAY

ChatGPT-4 can generate accurate treatment plans aligned with clinical practice guidelines, but can make mistakes when integrating multiple patient factors and understanding disease severity and progression.

ABSTRACT

Background: Artificial intelligence (AI) is engineered to emulate tasks that have historically required human interaction and intellect, including learning, pattern recognition, decision-making, and problem-solving. Although AI models like ChatGPT-4 have demonstrated satisfactory performance on medical licensing exams, suggesting a potential for supporting medical diagnostics and decision-making, no study of which we are aware has evaluated the ability of these tools to make treatment recommendations when given clinical vignettes and representative medical imaging of common orthopaedic conditions. As AI continues to advance, a thorough understanding of its strengths and limitations is necessary to inform safe and helpful integration into medical practice. Questions/purposes: (1) What is the concordance between ChatGPT-4-generated treatment recommendations for common orthopaedic conditions with both the American Academy of Orthopaedic Surgeons (AAOS) clinical practice guidelines (CPGs) and an orthopaedic attending physician's treatment plan? (2) In what specific areas do the ChatGPT-4-generated treatment recommendations diverge from the AAOS CPGs? Methods: Ten common orthopaedic conditions with associated AAOS CPGs were identified: carpal tunnel syndrome, distal radius fracture, glenohumeral joint osteoarthritis, rotator cuff injury, clavicle fracture, hip fracture, hip osteoarthritis, knee osteoarthritis, ACL injury, and acute Achilles rupture. For each condition, the medical records of 10 deidentified patients managed at our facility were used to construct clinical vignettes that each had an isolated, single diagnosis with adequate clarity. The vignettes also encompassed a range of diagnostic severity to evaluate more thoroughly adherence to the treatment guidelines outlined by the AAOS. These clinical vignettes were presented alongside representative radiographic imaging. The model was prompted to provide a single treatment plan recommendation. Each treatment plan was compared with established AAOS CPGs and to the treatment plan documented by the attending orthopaedic surgeon treating the specific patient. Vignettes where ChatGPT-4 recommendations diverged from CPGs were reviewed to identify patterns of error and summarized. Results: ChatGPT-4 provided treatment recommendations in accordance with the AAOS CPGs in 90% (90 of 100) of clinical vignettes. Concordance between ChatGPT-generated plans and the plan recommended by the treating orthopaedic attending physician was 78% (78 of 100). One hundred percent (30 of 30) of ChatGPT-4 recommendations for fracture vignettes and hip and knee arthritis vignettes matched with CPG recommendations, whereas the model struggled most with recommendations for carpal tunnel syndrome (3 of 10 instances demonstrated discordance). ChatGPT-4 recommendations diverged from AAOS CPGs for three carpal tunnel syndrome vignettes; two ACL injury,

rotator cuff injury, and glenohumeral joint osteoarthritis vignettes; as well as one acute Achilles rupture vignette. In these situations, ChatGPT-4 most often struggled to correctly interpret injury severity and progression, incorporate patient factors (such as lifestyle or comorbidities) into decision-making, and recognize a contraindication to surgery. Conclusion: ChatGPT-4 can generate accurate treatment plans aligned with CPGs but can also make mistakes when it is required to integrate multiple patient factors into decision-making and understand disease severity and progression. Physicians must critically assess the full clinical picture when using AI tools to support their decision-making. Clinical relevance: ChatGPT-4 may be used as an on-demand diagnostic companion, but patient-centered decision-making should continue to remain in the hands of the physician. Copyright © 2024 by the Association of Bone and Joint Surgeons.

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Abstract

Background

Artificial intelligence (AI) is engineered to emulate tasks that have historically required human interaction and intellect, including learning, pattern recognition, decision-making, and problem-solving. Although AI models like ChatGPT-4 have demonstrated satisfactory performance on medical licensing exams, suggesting a potential for supporting medical diagnostics and decision-making, no study of which we are aware has evaluated the ability of these tools to make treatment recommendations when given clinical vignettes and representative medical imaging of common orthopaedic conditions. As AI continues to advance, a thorough understanding of its strengths and limitations is necessary to inform safe and helpful integration into medical practice.

Questions/purposes

(1) What is the concordance between ChatGPT-4-generated treatment recommendations for common orthopaedic conditions with both the American Academy of Orthopaedic Surgeons (AAOS) clinical practice guidelines (CPGs) and an orthopaedic attending physician's treatment plan? (2) In what specific areas do the ChatGPT-4-generated treatment recommendations diverge from the AAOS CPGs?

Methods

Ten common orthopaedic conditions with associated AAOS CPGs were identified: carpal tunnel syndrome, distal radius fracture, glenohumeral joint osteoarthritis, rotator cuff injury, clavicle fracture, hip fracture, hip osteoarthritis, knee osteoarthritis, ACL injury, and acute Achilles rupture. For each condition, the medical records of 10 deidentified patients

managed at our facility were used to construct clinical vignettes that each had an isolated, single diagnosis with adequate clarity. The vignettes also encompassed a range of diagnostic severity to evaluate more thoroughly adherence to the treatment guidelines outlined by the AAOS. These clinical vignettes were presented alongside representative radiographic imaging. The model was prompted to provide a single treatment plan recommendation. Each treatment plan was compared with established AAOS CPGs and to the treatment plan documented by the attending orthopaedic surgeon treating the specific patient. Vignettes where ChatGPT-4 recommendations diverged from CPGs were reviewed to identify patterns of error and summarized.

Results

ChatGPT-4 provided treatment recommendations in accordance with the AAOS CPGs in 90% (90 of 100) of clinical vignettes. Concordance between ChatGPT-generated plans and the plan recommended by the treating orthopaedic attending physician was 78% (78 of 100). One hundred percent (30 of 30) of ChatGPT-4 recommendations for fracture vignettes and hip and knee arthritis vignettes matched with CPG recommendations, whereas the model struggled most with recommendations for carpal tunnel syndrome (3 of 10 instances demonstrated discordance). ChatGPT-4 recommendations diverged from AAOS CPGs for three carpal tunnel syndrome vignettes; two ACL injury, rotator cuff injury, and glenohumeral joint osteoarthritis vignettes; as well as one acute Achilles rupture vignette. In these situations, ChatGPT-4 most often struggled to correctly interpret injury severity and progression, incorporate patient factors (such as lifestyle or comorbidities) into decision-making, and recognize a contraindication to surgery.

Conclusion

ChatGPT-4 can generate accurate treatment plans aligned with CPGs but can also make mistakes when it is required to integrate multiple patient factors into decision-making and understand disease severity and progression. Physicians must critically assess the full clinical picture when using AI tools to support their decision-making.

Clinical Relevance

ChatGPT-4 may be used as an on-demand diagnostic companion, but patient-centered decision-making should continue to remain in the hands of the physician.

Introduction

Artificial intelligence (AI) is engineered to emulate tasks that typically necessitate human intellect, including learning, decision-making, and complex problem-solving. The field has progressed substantially since the 1960s, with recent breakthroughs in AI-driven chatbots [18]. These chatbots, such as ChatGPT, Google Gemini, and Claude, have earned public commendation for their ability to condense and collate available information, produce fluent conversation, respond to complex inquiries, and generate a variety of digital content [11,22,26]. Even within a specific chatbot, new versions demonstrate considerable improvements in their learning and enhancements, as seen between version 3 and version 4 of ChatGPT from 2020 to 2023. Despite the fact that they are seeing wider use in many settings, the swift adoption of AI chatbots across multiple industries has been met with both anticipation and skepticism, particularly concerning the reflection of historical biases and hallucinations in their outputs [28]. The utility of AI chatbots has been scrutinized across numerous professional fields, and they have demonstrated a level of competence by passing stringent exams like the Uniform Bar Exam and the United States Medical Licensing Exam [17,24]. Their performance in such tests suggests AI's potential for supporting medical diagnostics, documentation, and decision-making processes. When used to help create clinical visit documentation, ChatGPT created higher quality patient histories than physician dictations [14].

In orthopaedic surgery, a profession that heavily depends on medical imaging and clinical history for diagnosis and treatment, AI chatbots' analytical support could benefit healthcare professionals. As a result, these chatbots are being used more often to support clinical care in the absence of scientific evidence of their efficacy. Few orthopaedic studies exist that

analyze use cases for ChatGPT-4 in orthopaedic surgery. One investigation pitted ChatGPT-4's performance on orthopaedic assessments against that of orthopaedic residents [16,20,23], with noteworthy results in decision-making, diagnosis, and treatment recommendations for patients with lumbar spinal stenosis [27]. Another harnessed ChatGPT-4's capabilities to address patient queries regarding total hip, elbow, and shoulder arthroplasty, illustrating its expanding role in patient education and care [25,30]. However, ChatGPT-4's ability to generate accurate and comprehensive recommendations for the management of orthopaedic conditions is unknown. Although many use cases can be conceived, including assisting on-call residents with consideration of management options or acting as an "AI colleague" for an orthopaedic attending planning a case, it is crucial to develop a thorough understanding of its strengths and limitations to inform safe and helpful integration into medical practice.

We therefore asked: (1) What is the concordance between ChatGPT-4-generated treatment recommendations for common orthopaedic conditions with both the American Academy of Orthopaedic Surgeons (AAOS) clinical practice guidelines (CPGs) and an orthopaedic attending physician's treatment plan? (2) In what specific areas do the ChatGPT-4-generated treatment recommendations diverge from the AAOS CPGs?

Materials and Methods

We identified 10 common orthopaedic conditions that are covered in the established, evidence-based AAOS CPGs [10]. These conditions were as follows: carpal tunnel syndrome [2], distal radius fracture [3], glenohumeral joint osteoarthritis [4], rotator cuff injury [8], clavicle fracture [9], hip fracture [5], hip osteoarthritis [6], knee osteoarthritis [7], ACL injury [1], and acute Achilles rupture [19]. We chose these conditions to encompass a broad range of orthopaedic conditions so as to provide a comprehensive evaluation of ChatGPT-4's performance. For each condition, we constructed 10 clinical vignettes drawn from patients managed at our institution, a tertiary-care, urban Level 1 trauma center in Chicago, IL, USA. The patients were identified through a chart review of physicians' clinic schedules, and their protected health information was deidentified. We chose the clinical vignettes based on the following: diagnosis with an established and published AAOS CPG guideline and clinical documentation demonstrates an isolated, single diagnosis with adequate clarity. A clinical vignette was not included if the medical documentation did not provide the necessary patient information needed to arrive at the appropriate diagnosis and treatment as outlined by the AAOS CPG specific to that patient's pathology. Also, a broad range of disease severity was selected to mimic the stepwise and disease severity–specific treatment suggestions outlined by the AAOS guidelines.

We constructed clinical vignettes from the patient's initial history and clinical visit note. They included the history of present illness, medical history, surgical history, medications, social history, imaging interpretation, and physical exam. If the history and physical were found to provide insufficient information, additional content was added from the emergency medicine history and physical note to provide ChatGPT-4 with the necessary information to make the correct diagnosis. We stripped all protected health information and clinician identifying information from the vignettes to protect confidentiality. The assessment and plan section of each note were intentionally excluded to prevent biasing ChatGPT-4's recommendations. Patient imaging associated with the condition was not used because of confidentiality concerns. ChatGPT-4 was therefore provided with a characteristic radiograph or MRI image pathognomonic for the patient's pathology obtained from publicly available Google Images. After integration of the imaging and vignette, ChatGPT-4 was prompted to provide a single recommendation for treatment at the level of an orthopaedic attending. If it presented multiple treatment options, we continued to prompt the model until a single suggestion was generated. Prompt templates were used for both the initial and follow-up prompts to mitigate prompt-related variability (Supplemental Digital Content 1;http://links.lww.com/CORR/B327). In total, we presented 100 patient scenarios to ChatGPT-4, 10 clinical vignettes for each orthopaedic condition.

Primary and Secondary Study Outcomes

Our primary goal was to assess the concordance between ChatGPT-4-generated treatment recommendations, AAOS CPGs, and orthopaedic attending treatment plans. Since the AAOS guidelines have established clear-cut recommendations, we marked concordance of ChatGPT-4 responses with these recommendations as either “satisfactory” or “not satisfactory.” We marked concordance similarly between the documented treating orthopaedic attending’s management plan and the ChatGPT-4-generated recommendations. Then, we tallied the number of ChatGPT-4 recommendations that aligned with the AAOS CPGs and the orthopaedic attending treatment plan for each condition. Our secondary study goal was to identify areas in which ChatGPT-4-generated treatment recommendations diverged from the AAOS CPGs. We manually reviewed cases of divergence and qualitative reporting of errors in ChatGPT-4’s evaluation of the patient vignette and treatment options, which may explain its recommendation of a treatment plan that was not aligned with the CPGs.

Ethical Approval

We obtained ethical approval for this study from the institutional review board at The University of Chicago, Chicago, IL, USA (IRB23-0462).

Use of AI in This Work

We did not use AI or any LLM in any part of the process of writing or creation of this manuscript. LLM use was restricted to the experiment as we described it; it was not used in any way during the writing, editing, or revision of this manuscript.

Results

Concordance Between ChatGPT-4, AAOS CPGs, and the Physician’s Treatment Plan

ChatGPT-4 provided treatment recommendations in accordance with AAOS CPGs in 90% (90 of 100) of patient scenarios (Table1). The ChatGPT-4 recommendations matched the physician treatment plan proposed in 78% (78 of 100) of scenarios. All 30 ChatGPT recommendations for fracture vignettes as well as hip and knee arthritis vignettes were aligned with the AAOS CPGs. The model provided the highest proportion (3 of 10) of incorrect recommendations for carpal tunnel syndrome cases.

Proportion of treatment recommendations from ChatGPT-4 that aligned with AAOS CPGs and the treatment plan documented by the orthopaedic attending

Divergence of ChatGPT-4 and AAOS CPG Treatment Plans

ChatGPT-4 recommendations diverged from AAOS CPGs for 3 of 10 cases of carpal tunnel syndrome; 2 of 10 clinical vignettes of ACL injury, rotator cuff injury, and glenohumeral joint osteoarthritis; as well as 1 of 10 vignettes of acute Achilles rupture (Table2). For the three patients with carpal tunnel syndrome, ChatGPT-4 had difficulty deciding between nonoperative and surgical management, choosing nonoperative management in two patients with severe carpal tunnel (as determined by electromyography) and operative management in a patient with mild symptoms and an equivocal physical exam. In the two vignettes describing young, active patients with ACL tears, ChatGPT failed to recognize the patients’ activity levels as an indication for operative management. Furthermore, for a patient with glenohumeral joint osteoarthritis because of cuff tear arthropathy, ChatGPT-4 recommended rotator cuff repair, which is not in agreement with the AAOS CPG. In the two clinical vignettes of rotator cuff tears, ChatGPT-4 failed to consider the size and progression of the injury, recommending operative repair of a small rotator cuff tear that was improving with therapy and nonoperative management for a patient with a large, acute tear. Finally, in the vignettes of acute Achilles tendon tear, ChatGPT-4 suggested operative repair, failing to account for the increased surgical risk posed by the patient’s uncontrolled diabetes. There were also 22 clinical vignettes in which ChatGPT recommendations differed from the orthopaedic attending treatment plan (Table3). In some of these vignettes, both recommendations were acceptable under the AAOS CPGs.

Comparison of ChatGPT-4 recommendations with AAOS CPGs for all recommendations out of alignment with guidelines

Clinical scenarios in which the recommendations by ChatGPT-4 and the physician diverged

Discussion

AI-driven chatbots have seen wider use in medical practice recently, as they have showcased their ability to respond to complex inquiries and demonstrate complex problem-solving skills. Breakthroughs in this software could have vast applications in improving and supporting care for patients with orthopaedic conditions. As AI models become more heavily integrated into healthcare systems and decision-making, it is crucial that their accuracy is evaluated and that their strengths and weaknesses are characterized. This study aimed to assess ChatGPT-4's ability to discern common orthopaedic diagnoses from clinical presentations and compare ChatGPT-4's treatment recommendations for these diagnoses to evidence-based clinical practice guidelines and orthopaedic surgeon treatment plans. Overall, most of ChatGPT-4's treatment recommendations were in accordance with the AAOS CPG. Although our findings suggest that ChatGPT-4 can provide accurate and detailed suggestions for patient care, they also highlight its inability to consistently integrate patient factors into care plans and comprehend disease severity and progression. Although this study provides support for the use of ChatGPT-4 as an adjunct to clinical decision-making, clinician users must make their own independent decisions that factor in patient preferences, with an understanding that AI-generated recommendations may fail to encompass the full picture. Thus, the physician's responsibility remains paramount, with AI acting as a virtual colleague rather than a physician replacement.

Limitations

This study has several important limitations. Although an understanding of large language models (LLM) at their current level of development is important, the broader questions regarding continued improvements in modeling and knowledge are inevitable. The current study is solely a snapshot of the current available capabilities of LLM models. Even though we expect the accuracy of AI to naturally improve over time, our goal, more broadly, was to demonstrate current use cases for these tools to inform clinician users; identify areas in which LLMs struggle in making complex decisions; and demonstrate the importance of partnership with these tools to identify, advise, and improve adherence with evidence-based guidelines rather than replace physicians. As AI models continue to improve, it is likely that patient-centered decision-making and nuanced understanding of disease severity will continue to pose challenges. In addition, ChatGPT-4's response is entirely dependent on the specific prompt provided. Prompt engineering remains an important aspect of interacting with LLM models as small changes in the wording can result in considerable change to output. ChatGPT-4's treatment recommendations may change if given slight variations to the prompt provided. This was an important consideration during this study. However, the authors experimented with the prompt template to improve the generation of detailed and decisive responses. Furthermore, the weaknesses we have noted are in the model's inability to accommodate multiple factors, which is unlikely to be significantly mitigated through alternative prompting. Nevertheless, prompt engineering will remain a critical component of improving LLM applications in healthcare settings. The results of our study also depended on the quality of the orthopaedic surgeon–written clinical note. Although they introduced variability, these notes were used to most accurately capture the considerations being made by the orthopaedic surgeon. Furthermore, if the history of the present illness was found to be inadequate, it was supplemented using information from the emergency department professional's note. In addition, the use of representative images instead of patient imaging may have also influenced ChatGPT-4's recommended plan. This was done to protect patient confidentiality, and we chose images that closely matched the patient's pathology. As a result, it is unlikely that the use of representative (rather than each patient's actual) imaging would account for differences in ChatGPT-4's treatment plan.

Notably, there were several vignettes in which ChatGPT-4 and the physician recommendations did not match, although both remained in agreement with the CPGs. This discrepancy is possible because the CPGs often provide more than one appropriate treatment option for a given condition. For example, when a patient sustains a valgus impacted femoral neck

fracture, CPGs state that multiple treatment options are acceptable: “Stable femoral neck fractures: In patients with stable (impacted/nondisplaced) femoral neck fractures, hemiarthroplasty, internal fixation or non-operative care may be considered” [5]. Also, for displaced femoral neck fractures, both hemiarthroplasty and THA are acceptable treatments: “Unstable femoral neck fractures—total arthroplasty vs hemi arthroplasty: In properly selected patients with unstable (displaced) femoral neck fractures, there may be a functional benefit to total hip arthroplasty over hemi arthroplasty at the risk of increasing complications” [5]. This CPG leaves it up to the discretion of the surgeon to determine hemiarthroplasty versus THA. This flexibility creates instances in which both ChatGPT-4 and physician recommendations were satisfactory but not in alignment, highlighting the presence of nuances in the decision-making process that only the physician can thoroughly discern and individually apply to each patient’s situation. Thus, the physician should continue to be at the center of patient-physician decision-making, but they can use AI as treatment adjuvants rather than physician replacements.

Although choosing conditions associated with CPGs provided an objective standard for comparison, it did not allow us to capture ChatGPT-4’s performance for injuries with less consensus on best practices for management. Evaluation of AI management of these conditions will require more sophisticated methods of determining satisfactory recommendations. However, analysis of ChatGPT-4 performance on injuries in which management is more standardized acts as a reasonable start in understanding its aptitude. Additionally, the clinical vignettes were drawn from patients at a single institution and may not comprehensively test ChatGPT-4’s aptitude in managing a wide variety of presentations of a single condition. However, the vignettes were drawn from a variety of orthopaedic attendings, and we ensured that there was a diversity of disease severity and presentation among the 10 vignettes for each condition.

Concordance Between ChatGPT-4, AAOS CPGs, and the Physician’s Treatment Plan

ChatGPT-4 accurately provided recommendations across all 10 orthopaedic conditions, with 90 of 100 recommendations in accordance with AAOS CPGs. This suggests that, when given access to relevant patient information and imaging, AI can play a supportive role in suggesting evidence-based treatment pathways. In addition to providing recommendations that aligned with CPGs in most clinical vignettes, ChatGPT-4 also commented on some of the technical nuances of surgery. When describing a treatment plan for a displaced hip fracture in an active patient (a 65-year-old man), ChatGPT not only correctly recommended a THA, but also recommended a cemented femoral and press-fit acetabular component [15]. When describing percutaneous pinning for a nondisplaced femoral neck fracture, it correctly suggested the inverted triangle configuration when placing screws [12]. It also appropriately described how a sliding hip screw was not an appropriate construct for an unstable (reverse oblique) intertrochanteric femur fracture and instead stated a cephalomedullary nail was the preferred implant [21,29]. For ACL reconstructions, it suggested using either a hamstring autograft or a patella tendon autograft [13]. These recommended surgical techniques for these various orthopaedic injuries are in accordance with the evidence and are a testament to the potential of AI to generate detailed surgical plans. Further research may help characterize the ability of newer AI models to make recommendations for these conditions as well as injuries without established practice guidelines.

Divergence of ChatGPT-4 and AAOS CPG Treatment Plans

The 10 clinical vignettes where ChatGPT-4’s recommendations diverged from the AAOS CPGs offer valuable insights into the current limitations and areas for improvement of AI in clinical settings. In these cases, ChatGPT-4 most often struggled to correctly interpret injury severity and progression to decide between operative and nonoperative treatment. It also had difficulty incorporating other patient factors, including lifestyle and comorbidities, to analyze operative risk and determine an appropriate treatment course. In one vignette, ChatGPT-4 even failed to recognize a contraindication and inappropriately recommended surgery. As a result, it is critical that clinicians who use ChatGPT-4 for decision-making understand that its recommendations may fail to account for patient factors and that it even has the potential to recommend contraindicated interventions. We encourage those who are using this tool to make their own independent patient-care decisions using ChatGPT-4 as an AI colleague. Further research may elucidate prompt factors, such as the length and

structure of the prompt, that facilitate generation of the most accurate and useful clinical recommendations. The aptitude of newer AI models and their ability to weigh multiple factors may also be investigated.

Conclusion

The current study has demonstrated the ability of LLM and AI, specifically ChatGPT-4, to generate accurate and comprehensive treatment plans based on available evidence-based information. However, it also revealed the shortcomings of this technology, which had difficulty integrating multiple patient factors into decision-making and understanding disease severity and progression. Our findings suggest that AI may be useful as a virtual colleague, providing trainees with a discussion of how the case relates to clinical practice guidelines or double-checking a physician's reasoning. However, physicians must ultimately rely on their own critical thinking to factor nuances into the decision-making process. Future investigation aimed at understanding and improving AI-aided decision-making has the potential to guide AI integration into orthopaedic practice. This may involve experimentation on prompt engineering frameworks that improve ChatGPT-4's decision-making capabilities. For example, in our trial of different prompts, we noted that asking ChatGPT to draw from existing guidelines generated higher quality responses. In the future, quantitative analyses of different prompt lengths and structures would provide clinicians with guidance on how to best utilize AI tools. Additionally, ChatGPT-4's ability to provide recommendations for more complicated clinical vignettes regarding conditions without established guidelines would be useful in better characterizing ChatGPT-4's utility in a broader set of clinical scenarios. Finally, a comparison of recommendations generated by a wider set of LLMs, including Gemini and Claude, would assist clinicians in deciding which tool to use in their practices [11,26].

Supplementary Material

Footnotes

Each author certifies that there are no funding or commercial associations (consultancies, stock ownership, equity interest, patent/licensing arrangements, etc.) that might pose a conflict of interest in connection with the submitted article related to the author or any immediate family members.

All ICMJE Conflict of Interest Forms for authors andClinical Orthopaedics and Related Research®editors and board members are on file with the publication and can be viewed on request.

Ethical approval for this study was obtained from the institutional review board at The University of Chicago, Chicago, IL, USA (IRB23-0462).

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References

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